Block practice · Trigonometric identities and equations

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Deriving the double-angle formulae

Solving trig equations with a multiple or shifted angle

Proving identities with double-angle and addition formulae

OLevels.com SCHOOL

Practice workbook · Deriving the double-angle formulae

Getting sin 2A, cos 2A (in all three forms) and tan 2A from the addition formulae by putting B = A.

24 question parts: 1 worked-example retry and 23 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-TRI-20

1. Question 11 · part a · 3 marks

2021 JUN · Paper 01

Worked-example retry · Attempt part a only.

11 (a) Using a formula from page 2, show that $\cos 2x = 1 - 2\sin^2 x$

(3)

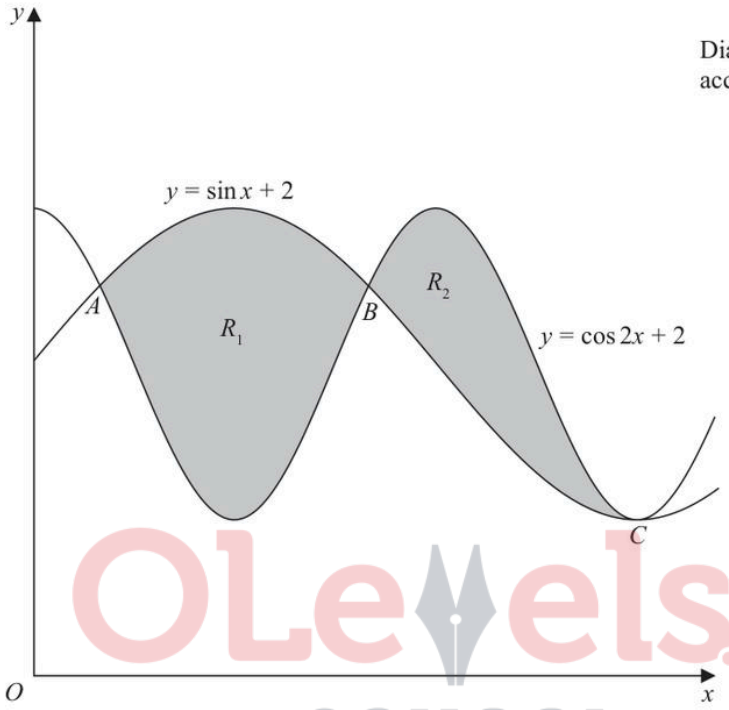


Figure 2

Figure 2 shows a sketch of part of the curves with equations $y = \sin x + 2$ and $y = \cos 2x + 2$

The points A, B and C, shown in Figure 2, are three points that are common to both curves.

(b) Find the coordinates of each of these points.

(4)

R_1 and R_2, shown shaded in Figure 2, are two regions enclosed by the two curves.

(c) Use calculus to find, in its simplest form, the ratio

area of R_1 : area of R_2

(8)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

P 6 6 0 2 4 A 0 3 2 3 6

1. Question 11 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 10 · part a · 2 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

10 (a) Use the formula for $\cos(A + B)$ to show that $\cos 2A = 2\cos^2 A - 1$

(2)

(b) Show that $\cos 4A = 8\cos^4 A - 8\cos^2 A + 1$

(4)

(c) Solve the equation $\cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right)\left[\cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right) - 1\right] = -\frac{1}{16} \qquad 0 \leqslant \theta < 2\pi$

Give your answers in terms of π.

(5)

$$f(A) = 4\cos^4 A - 4\cos^2 A + 1$$

(d) Using calculus, find the exact value of $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} f(A)\,dA$

Give your answer in the form $a\pi - b\sqrt{c}$ where a and b are fractions in their lowest terms and c is a prime number.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

2. Question 10 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 8 · part c · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part c only.

8 (a) Given that x is measured in radians, complete the table of values for

$$y = \sin x - 3\cos 2x - 0.5$$

Give your answers to one decimal place.

x	0	0.5	0.8	1	1.6	2	2.5	3
y			0.3	1.6			–0.8	–3.2

(2)

(b) On the grid opposite, draw the graph of $y = \sin x - 3\cos 2x - 0.5$ for $0 \leqslant x \leqslant 3$

(2)

(c) Use a formula from page 2 to show that $\cos 2A = 1 - 2\sin^2 A$

(2)

$$f(x) = 2\sin x + 12\sin^2 x - x - 5$$

(d) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of the roots of the equation $f(x) = 0$ in the interval $0 \leqslant x \leqslant 3$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

3. Question 8 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 10 · part a.i · 0 marks

2023 JAN · Paper 02

Further practice · Attempt part a.i only.

10 Using formulae given on page 2

(a) show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

Given that $\theta \neq (90° + 180°n)$ where $n \in \mathbb{Z}$

(b) use the results from part (a) to show that $\sin 2\theta - \tan\theta$ can be written as $\tan\theta\cos 2\theta$

(4)

(c) Solve for $0 < x < 360$

$$\sin 2x° - \tan x° = 0$$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

4. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 10 · part a.ii · 5 marks

2023 JAN · Paper 02

Further practice · Attempt part a.ii only.

10 Using formulae given on page 2

(a) show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

Given that $\theta \neq (90° + 180°n)$ where $n \in \mathbb{Z}$

(b) use the results from part (a) to show that $\sin 2\theta - \tan\theta$ can be written as $\tan\theta\cos 2\theta$

(4)

(c) Solve for $0 < x < 360$

$$\sin 2x° - \tan x° = 0$$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

5. Question 10 · part a.ii · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 9 · part a.i · 0 marks

2023 JUN · Paper 01

Further practice · Attempt part a.i only.

9 (a) Using the formulae on page 2, show that

(i) $\cos^2 A = \dfrac{\cos 2A + 1}{2}$

(ii) $\sin^2 A = \dfrac{1 - \cos 2A}{2}$

(4)

(b) Show that

$$(2\sin x - \cos x)(\sin x - 3\cos x) = \frac{1}{2}(\cos 2x - 7\sin 2x + 5)$$

(5)

$$y = (2\sin x - \cos x)(\sin x - 3\cos x)$$

(c) Solve, for $0° \leqslant x \leqslant 180°$ the equation, $\dfrac{dy}{dx} = 0$

Give your answers to the nearest whole number.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

6. Question 9 · part a.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 9 · part a.ii · 4 marks

2023 JUN · Paper 01

Further practice · Attempt part a.ii only.

9 (a) Using the formulae on page 2, show that

(i) $\cos^2 A = \dfrac{\cos 2A + 1}{2}$

(ii) $\sin^2 A = \dfrac{1 - \cos 2A}{2}$

(4)

(b) Show that

$$(2\sin x - \cos x)(\sin x - 3\cos x) = \frac{1}{2}(\cos 2x - 7\sin 2x + 5)$$

(5)

$$y = (2\sin x - \cos x)(\sin x - 3\cos x)$$

(c) Solve, for $0° \leqslant x \leqslant 180°$ the equation, $\dfrac{dy}{dx} = 0$

Give your answers to the nearest whole number.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 9 · part a.ii · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 11 · part a · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

11 (a) Using a formula on page 2 show that $\cos 2A = 2\cos^2 A - 1$

(2)

(b) Hence show that $\left(2\cos^2 A - 1\right)^2 = \dfrac{\cos 4A + 1}{2}$

(3)

The curve with equation $y = \dfrac{\sin 2x}{2} + \dfrac{\left(2\cos^2 x - 1\right)^2}{2} + \dfrac{1}{8}$ has a stationary point P in the range $0 \leqslant x \leqslant \dfrac{\pi}{6}$

(c) Find the exact coordinates of P

(7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8. Question 11 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 11 · part a.i · 3 marks

2024 JUN · Paper 02

Further practice · Attempt part a.i only.

11 Using formulae from page 2, show that

(a) (i) $\cos 2A = 2\cos^2 A - 1$

(3)

(ii) $\sin 2A = 2\sin A \cos A$

(1)

(b) Show that $\cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$

(4)

Hence, or otherwise,

(c) solve, giving exact values in terms of π

$$8\cos^3\left(\frac{\theta}{2}\right) - 6\cos\left(\frac{\theta}{2}\right) - 1 = 0 \qquad \text{for } 0 \leqslant \theta \leqslant 2\pi$$

(4)

(d) use algebraic integration to find the exact value of

$$\int_0^{\frac{\pi}{6}} \left(4\cos^3\theta - \sin 2\theta\right) \mathrm{d}\theta$$

(4)

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

9. Question 11 · part a.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 11 · part a.ii · 1 mark

2024 JUN · Paper 02

Further practice · Attempt part a.ii only.

11 Using formulae from page 2, show that

(a) (i) $\cos 2A = 2\cos^2 A - 1$ (3)

(ii) $\sin 2A = 2\sin A \cos A$ (1)

(b) Show that $\cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$ (4)

Hence, or otherwise,

(c) solve, giving exact values in terms of π

$$8\cos^3\left(\frac{\theta}{2}\right) - 6\cos\left(\frac{\theta}{2}\right) - 1 = 0 \qquad \text{for } 0 \leqslant \theta \leqslant 2\pi$$ (4)

(d) use algebraic integration to find the exact value of

$$\int_0^{\frac{\pi}{6}} \left(4\cos^3\theta - \sin 2\theta\right) \mathrm{d}\theta$$ (4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10. Question 11 · part a.ii · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 8 · part a · 1 mark

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

8 (a) Write down the value of k such that $\sin 2A = k \sin A \cos A$

(1)

$$g(A) = 2 + 3\cos A - \sin A - 3\sin 2A - 2\cos^2 A$$

Given that $g(A)$ can be written in the form $(p\cos A - \sin A)(q - r\sin A)$ where p, q and r are integers,

(b) find the value of p, the value of q and the value of r.

(3)

(c) Hence solve, in radians to 3 significant figures where appropriate, the equation

$$g(2\theta) = 0 \quad \text{for} \quad 0 \leqslant \theta < \pi$$

(6)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

11. Question 8 · part a · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 1 · part a · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

1 (a) Use the formula for $\cos(A + B)$ to show that $\cos 2A = 1 - 2\sin^2 A$

(2)

Given that $\sin A = \frac{x+1}{2}$ and $\cos 2A = \frac{4-y}{3}$

(b) show that $y = \frac{1}{2}(3x^2 + 6x + 5)$

(3)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4

P 6 6 0 2 7 R R A 0 4 3 6

12. Question 1 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 6 · part a · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

6 (a) Using a formula from page 2, show that $\cos 2A = 1 - 2\sin^2 A$

(2)

The finite region R is bounded by the curve with equation $y = 3 + 2\sin x$, the x-axis, the y-axis and the line with equation $x = \frac{\pi}{4}$

The region R is rotated through 360° about the x-axis.

(b) Use calculus to find the volume of the solid generated.
Give your answer to the nearest integer.

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

13. Question 6 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 7 · part a · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

7 (i) (a) Using a formula from page 2, show that

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

(2)

Given that $\tan 2\alpha = 1$

(b) show that $\tan\alpha = a \pm \sqrt{b}$ where a and b are integers whose values need to be found.

(3)

(ii) (a) Using formulae from page 2, show that $\cos(x-30)^\circ = \sin(x+30)^\circ$ can be written as $\tan x^\circ = 1$

(4)

(b) Hence, or otherwise, solve

$$\cos(2y-30)^\circ = \sin(2y+30)^\circ \qquad \text{for } -90 < y \leqslant 90$$

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

14. Question 7 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

15. Question 9 · part a · 2 marks

2024 NOV · Paper 02

Further practice · Attempt part a only.

9 (a) Using a formula given on page 2, show that

$$\cos 2\theta = 2\cos^2\theta - 1$$

(2)

(b) Hence show that

$$\int_{\frac{\pi}{3}}^{\frac{3\pi}{4}} \left(2\cos^2\theta - 1\right) \mathrm{d}\theta = -\frac{a+\sqrt{b}}{c}$$

where a, b and c are integers to be found.

(4)

Diagram **NOT** accurately drawn

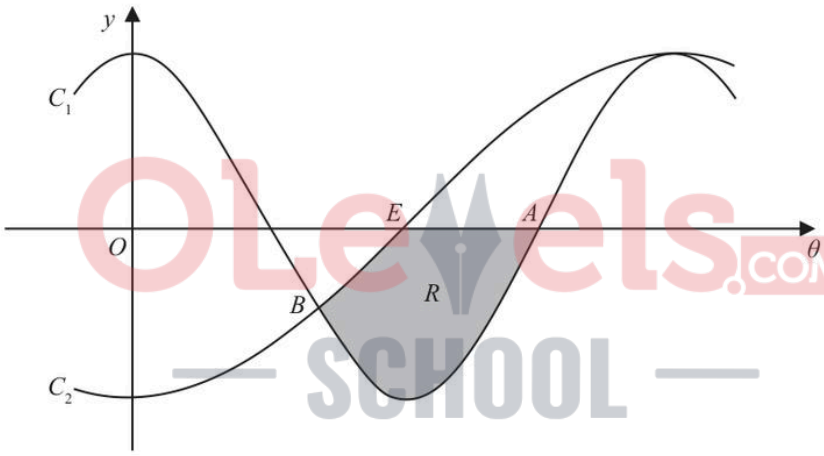


Figure 3

Figure 3 shows part of the curve C_1 with equation $y = 2\cos^2\theta - 1$ and part of the curve C_2 with equation $y = -\cos\theta$

Point B is the intersection of C_1 and C_2 as shown in Figure 3

Point $A\left(\frac{3\pi}{4}, 0\right)$ is the intersection of C_1 with the θ-axis as shown in Figure 3

Point $E\left(\frac{\pi}{2}, 0\right)$ is the intersection of C_2 with the θ-axis as shown in Figure 3

The finite region R, shown shaded in Figure 3, is bounded by the θ-axis, C_1 and C_2

(c) Use calculus to find, in its simplest form, the exact area of R

(8)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 3 5 8 7 A 0 2 4 3 2

15. Question 9 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

16. Question 11 · part a · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

11 (a) Use a formula on page 2 to show that $\sin^2 A = \frac{1}{2}(1 - \cos 2A)$

(3)

(b) Show that $\sin^4 x + \cos^4 x = \frac{3 + \cos 4x}{4}$

(5)

(c) Hence solve, in degrees to one decimal place, the equation

$$8\sin^4\left(\frac{\theta}{2}\right) + 8\cos^4\left(\frac{\theta}{2}\right) = 5\sin(2\theta) + 6 \quad \text{for} \quad 0^\circ \leqslant \theta < 180^\circ$$

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

16. Question 11 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

17. Question 10 · part b.i · 0 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b.i only.

10 (a) Use the factor theorem to show that $(4x - 3)$ is a factor of

$$16x^3 + 11x - 15$$

(2)

(b) Using formulae given on page 2, show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

(c) Show that the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

becomes the equation

$$16x^3 + 11x - 15 = 0$$

by using the substitution $x = \cos\theta$

(4)

(d) Hence show that any solution of the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

is given by $\cos\theta = \frac{3}{4}$

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

17. Question 10 · part b.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

18. Question 10 · part b.ii · 5 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b.ii only.

10 (a) Use the factor theorem to show that $(4x - 3)$ is a factor of

$$16x^3 + 11x - 15$$

(2)

(b) Using formulae given on page 2, show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

(c) Show that the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

becomes the equation

$$16x^3 + 11x - 15 = 0$$

by using the substitution $x = \cos\theta$

(4)

(d) Hence show that any solution of the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

is given by $\cos\theta = \frac{3}{4}$

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

18. Question 10 · part b.ii · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

19. Question 10 · part a.i · 0 marks

2023 NOV · Paper 01

Further practice · Attempt part a.i only.

10 (a) Using formulae on page 2, show that

(i) $\sin 2A = 2\sin A\cos A$

(ii) $\cos 2A = 2\cos^2 A - 1$

(3)

$$f(\theta) = \frac{2\tan\theta}{1+\tan^2\theta}$$

(b) Show that $f(\theta) = \sin 2\theta$

(4)

(c) Solve, in radians to 3 significant figures, for $-\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$, the equation

$$5\tan\left(x+\frac{\pi}{6}\right) = \left[1+\tan^2\left(x+\frac{\pi}{6}\right)\right]\left[1-2\cos^2\left(x+\frac{\pi}{6}\right)\right]$$

(6)

(d) Using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \left(\frac{4\tan\theta}{1+\tan^2\theta} - \cos 5\theta + 2\right) d\theta$$

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

19. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

20. Question 10 · part a.ii · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part a.ii only.

10 (a) Using formulae on page 2, show that

(i) $\sin 2A = 2\sin A\cos A$

(ii) $\cos 2A = 2\cos^2 A - 1$

(3)

$$f(\theta) = \frac{2\tan\theta}{1+\tan^2\theta}$$

(b) Show that $f(\theta) = \sin 2\theta$

(4)

(c) Solve, in radians to 3 significant figures, for $-\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$, the equation

$$5\tan\left(x+\frac{\pi}{6}\right) = \left[1+\tan^2\left(x+\frac{\pi}{6}\right)\right]\left[1-2\cos^2\left(x+\frac{\pi}{6}\right)\right]$$

(6)

(d) Using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \left(\frac{4\tan\theta}{1+\tan^2\theta} - \cos 5\theta + 2\right) d\theta$$

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

20. Question 10 · part a.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

21. Question 10 · part a.i · 0 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part a.i only.

10 (a) Using a formula from page 2, show that

(i) $\sin^2 A = \frac{1-\cos 2A}{2}$

(ii) $\cos^2 A = \frac{\cos 2A+1}{2}$

(5)

Diagram **NOT** accurately drawn

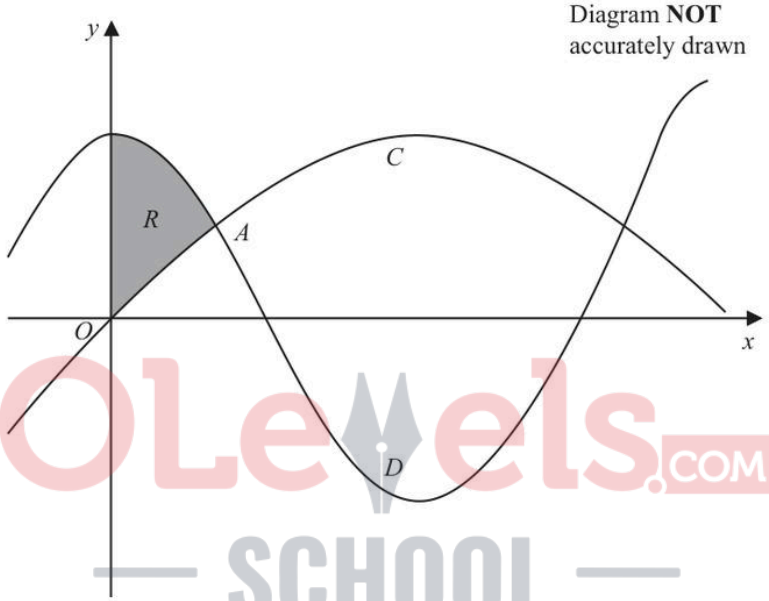


Figure 2

Figure 2 shows part of the curve C with equation $y = \sin x$ and part of the curve D with equation $y = \cos 2x$
Curve C and curve D intersect at the point A

(b) Use algebra to find the exact coordinates of A

(5)

The shaded region R is rotated through $360°$ about the x-axis.

(c) Use algebraic integration to find the exact volume of the solid generated.

Give your answer in the form $\frac{a\sqrt{a}}{b}\pi$

where a is a prime number and b is an integer.

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 6 5 0 7 A 0 2 6 3 2

21. Question 10 · part a.i · 0 marks · Working and answer

Working and answer

OLevels.com SCHOOL

22. Question 10 · part a.ii · 5 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part a.ii only.

10 (a) Using a formula from page 2, show that

(i) $\sin^2 A = \frac{1-\cos 2A}{2}$

(ii) $\cos^2 A = \frac{\cos 2A+1}{2}$

(5)

Diagram **NOT** accurately drawn

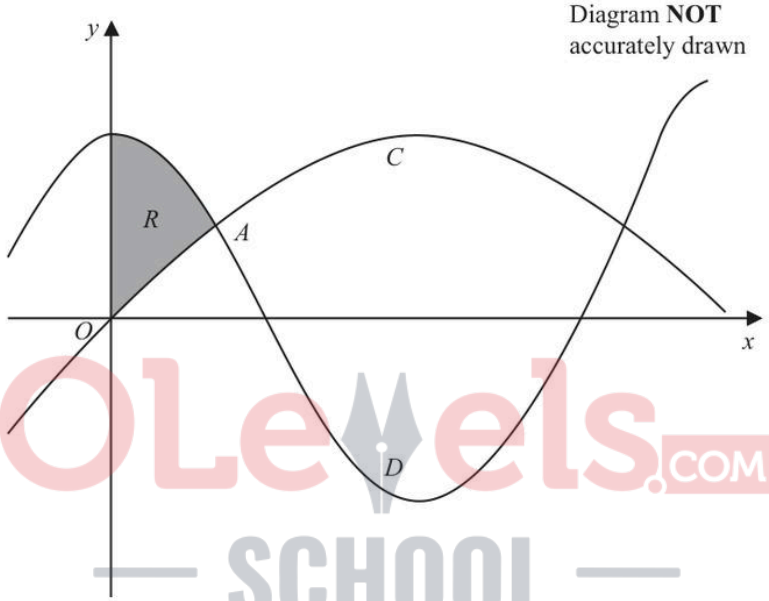


Figure 2

Figure 2 shows part of the curve C with equation $y = \sin x$ and part of the curve D with equation $y = \cos 2x$
Curve C and curve D intersect at the point A

(b) Use algebra to find the exact coordinates of A

(5)

The shaded region R is rotated through $360°$ about the x-axis.

(c) Use algebraic integration to find the exact volume of the solid generated.

Give your answer in the form $\frac{a\sqrt{a}}{b}\pi$

where a is a prime number and b is an integer.

(5)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 7 6 5 0 7 A 0 2 6 3 2

22. Question 10 · part a.ii · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

23. Question 11 · part a · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part a only.

11 (a) Using a formula on page 2, show that

$$\frac{1-\cos 2A}{1+\cos 2A} = \tan^2 A$$

(3)

(b) Hence, or otherwise, solve in degrees to one decimal place

$$\frac{3-3\cos 4x}{1+\cos 4x} + \frac{5\sin 2x}{\cos 2x} = 2 \quad \text{for } -90^\circ < x < 90^\circ$$

(7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

23. Question 11 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

24. Question 8 · part a · 2 marks

2024 NOV · Paper 01

Further practice · Attempt part a only.

8 (i) (a) Using a formula given on page 2, show that

$$\tan 2A = \frac{2\tan A}{1-\tan^2 A}$$

(2)

(b) Hence, solve the equation

$$\tan A° - \tan 2A° = 0 \qquad \text{for } 0 \leqslant A \leqslant 180$$

(5)

(ii) Using a formula given on page 2, solve, giving your solutions as exact values

$$\cos\left(x - \frac{\pi}{6}\right) = \sin x \qquad \text{for } -\pi \leqslant x \leqslant 2\pi$$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

24. Question 8 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Solving trig equations with a multiple or shifted angle

Solving sin, cos or tan of a single, multiple or shifted angle equal to a constant over a given interval.

13 question parts: 1 worked-example retry and 12 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-TRI-24

1. Question 8 · part a · 3 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part a only.

8 (a) Solve, to the nearest integer, the equation

$$\sin(2x - 75)° = -0.515 \quad \text{for } 0 \leqslant x < 180$$

(3)

(b) Giving your solutions to one decimal place, where appropriate, solve the equation

$$2\tan y° + 5\sin y° = 0 \quad \text{for } 0 \leqslant y \leqslant 180$$

(4)

(c) Explain mathematically why there are no values of θ that satisfy the equation

$$3\cos^2\theta° - 3\sin^2\theta° + \sin\theta° + 12 = 0$$

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

1. Question 8 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 10 · part c · 5 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

10 (a) Use the formula for $\cos(A + B)$ to show that $\cos 2A = 2\cos^2 A - 1$

(2)

(b) Show that $\cos 4A = 8\cos^4 A - 8\cos^2 A + 1$

(4)

(c) Solve the equation $\cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right)\left[\cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right) - 1\right] = -\frac{1}{16} \qquad 0 \leqslant \theta < 2\pi$

Give your answers in terms of π.

(5)

$$f(A) = 4\cos^4 A - 4\cos^2 A + 1$$

(d) Using calculus, find the exact value of $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} f(A)\,dA$

Give your answer in the form $a\pi - b\sqrt{c}$ where a and b are fractions in their lowest terms and c is a prime number.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

P 6 1 8 8 3 A 0 3 0 3 6

2. Question 10 · part c · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 7 · part b · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

7 Given that

$$\frac{3+\sin^2\theta}{\cos\theta-2}=3\cos\theta$$

(a) show that $\cos\theta=-\frac{1}{2}$

(4)

(b) Hence solve the equation

$$\frac{3+\sin^2 3x}{\cos 3x-2}=3\cos 3x \quad \text{for } 0^\circ \leqslant x < 180^\circ$$

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

3. Question 7 · part b · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 7 is 8 marks)

3. Question 7 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 10 · part a · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part a only.

10 (a) Solve the equation

$$\tan x^\circ = -3 \qquad \text{for } 0 \leqslant x < 360$$

Give your solutions to the nearest whole number.

(3)

Given that

$$7\sin^2\theta + \sin\theta\cos\theta = 6$$

(b) show that

$$\tan^2\theta + \tan\theta - 6 = 0$$

(3)

(c) Hence solve the equation

$$7\sin^2 y^\circ + \sin y^\circ \cos y^\circ = 6 \qquad \text{for } 0 \leqslant y < 360$$

Give your solutions to the nearest whole number.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

4. Question 10 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 10 · part a · 3 marks

2021 NOV · Paper 01

Further practice · Attempt part a only.

10

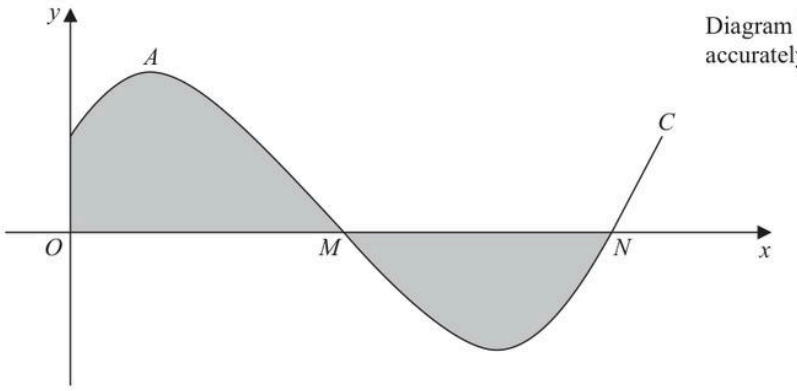


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows the curve C with equation $y = \frac{1}{2} + \sin 3x$ where $0 \leqslant x < \frac{2\pi}{3}$

The curve C crosses the x-axis at the points M and N

(a) Show that the coordinates of M are $\left(\frac{7\pi}{18}, 0\right)$ and find the coordinates of N

(3)

The curve C has a maximum at the point A

(b) Find the coordinates of A

(4)

(c) Find an equation of the tangent to C at M

Give your answer in the form $ay + b\sqrt{3}x - c\sqrt{3}\pi = 0$ where a, b and c are integers to be found.

(4)

The finite region, shown shaded in Figure 4, is bounded by the curve C, the y-axis and the part of the x-axis from O to N

(d) Use algebraic integration to find, to 3 significant figures, the total area of the shaded region.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 6 6 0 2 6 R A 0 2 6 3 2

5. Question 10 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 7 · part a · 4 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

7 (a) Solve the equation

$$\cos(3x - 15)° = \frac{\sqrt{3}}{2} \quad \text{for} \quad 0 \leqslant x < 180$$

(4)

(b) Solve, giving your solutions to one decimal place where appropriate,

$$3\tan y° + 4\sin y° = 0 \quad \text{for} \quad -180 \leqslant y < 180$$

(4)

(c) Solve, giving your solutions to one decimal place where appropriate,

$$\cos\theta° = 3\sin^2\theta° - 1 \quad \text{for} \quad -180 \leqslant \theta < 180$$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

6. Question 7 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 1 · part c · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

DO NOT WRITE IN THI

1 Here is a formula

$$P = 3 + 2\sin\left(\frac{3\pi t}{8}\right) \qquad 0 \leqslant t \leqslant 12$$

(a) Find the exact value of P when $t = \frac{10}{3}$

(2)

(b) Find

(i) the largest value of P

(ii) the smallest value of P

(2)

DO NOT WRITE IN THIS AREA

(c) Find the least value of t for which $P = 4$

(3)

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA

7. Question 1 · part c · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 1 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 1 · part c · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 1 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

(Total for Question 1 is 7 marks)

7. Question 1 · part c · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 10 · part a · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

10 (a) Solve the equation

$$\sin\left(x+\frac{\pi}{3}\right)=\frac{\sqrt{3}}{2} \qquad \text{for } 0 \leqslant x \leqslant 2\pi$$

Give your solutions in terms of π, where appropriate.

(3)

(b) Solve the equation

$$3\sin\theta + 5\cos\theta = 0 \qquad \text{for } -360° \leqslant \theta \leqslant 360°$$

Give your solutions to the nearest degree.

(3)

(c) Solve the equation

$$1 + \sin 2y = 2\cos^2 2y \qquad \text{for } -180° \leqslant y \leqslant 0°$$

(5)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8. Question 10 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 7 · part a · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

7

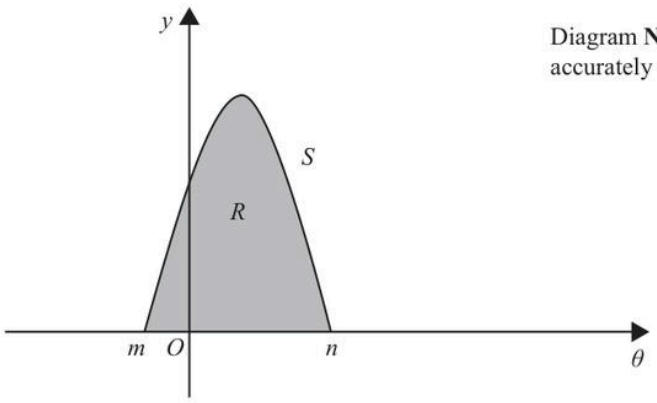


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows part of the curve S with equation $y = \left(\cos 3\theta + \sqrt{3}\sin 3\theta\right)^{\frac{1}{2}}$

where $m \leqslant \theta \leqslant n$

The curve S meets the x-axis at the point with coordinates $(m, 0)$ and at the point with coordinates $(n, 0)$

(a) Find the exact value of m and the exact value of n

(3)

The finite region R, shown shaded in Figure 2, is bounded by the curve S, and the x-axis in the region $m \leqslant \theta \leqslant n$

The region R is rotated through 2π radians about the theta-axis.

(b) Use calculus to find the exact volume of the solid generated.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

9. Question 7 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 7 · part b · 3 marks

2022 JUN · Paper 02

Further practice · Attempt part b only.

7 (i) (a) Using a formula from page 2, show that

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

(2)

Given that $\tan 2\alpha = 1$

(b) show that $\tan\alpha = a \pm \sqrt{b}$ where a and b are integers whose values need to be found.

(3)

(ii) (a) Using formulae from page 2, show that $\cos(x-30)^\circ = \sin(x+30)^\circ$ can be written as $\tan x^\circ = 1$

(4)

(b) Hence, or otherwise, solve

$$\cos(2y-30)^\circ = \sin(2y+30)^\circ \qquad \text{for } -90 < y \leqslant 90$$

(2)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10. Question 7 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 11 · part c · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

11 (a) Use a formula on page 2 to show that $\sin^2 A = \frac{1}{2}(1 - \cos 2A)$

(3)

(b) Show that $\sin^4 x + \cos^4 x = \frac{3 + \cos 4x}{4}$

(5)

(c) Hence solve, in degrees to one decimal place, the equation

$$8\sin^4\left(\frac{\theta}{2}\right) + 8\cos^4\left(\frac{\theta}{2}\right) = 5\sin(2\theta) + 6 \quad \text{for} \quad 0^\circ \leqslant \theta < 180^\circ$$

(4)

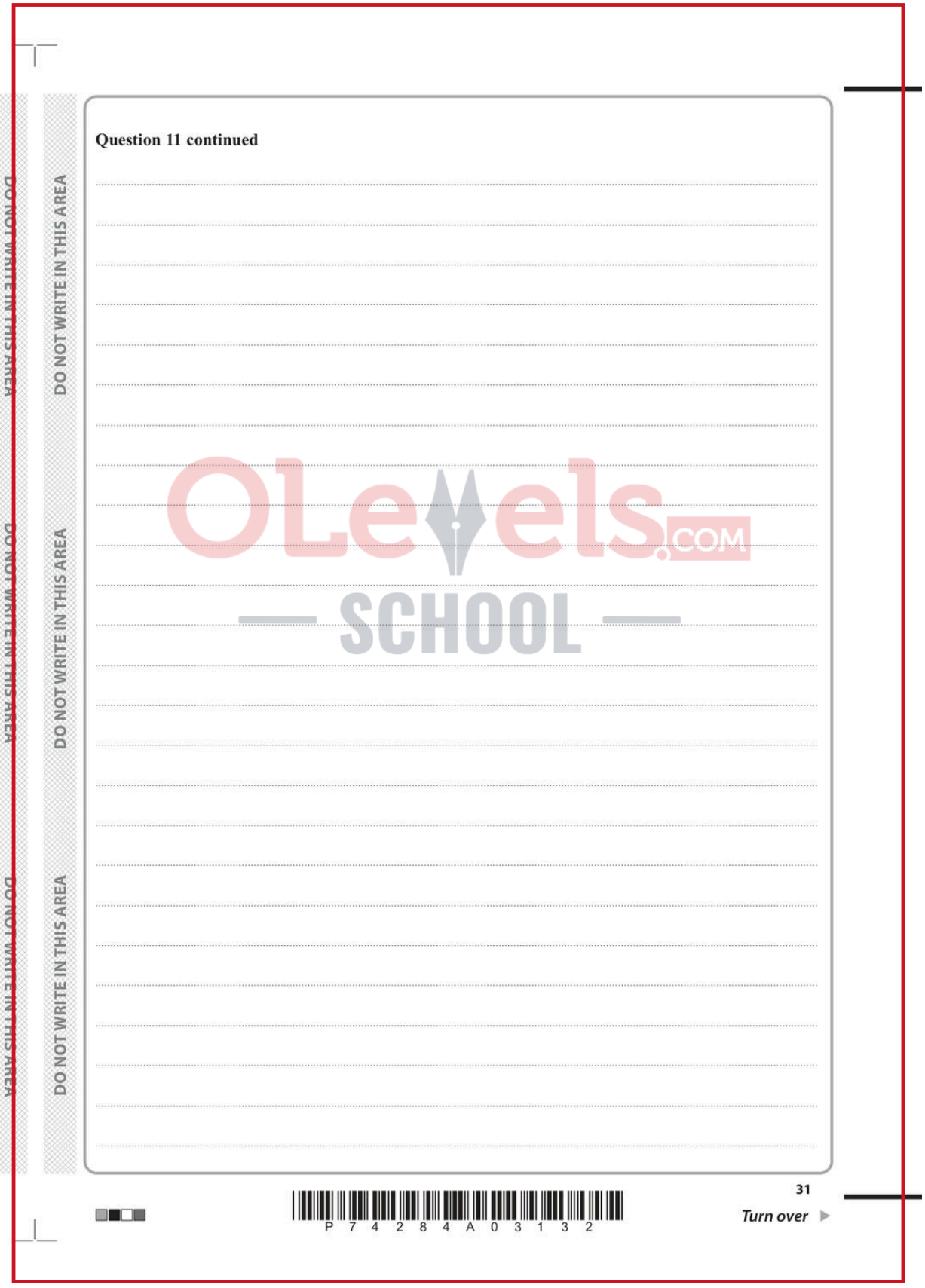
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

11. Question 11 · part c · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.



Question 11 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

11. Question 11 · part c · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 11 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

(Total for Question 11 is 12 marks)

11. Question 11 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 10 · part c · 6 marks

2023 NOV · Paper 01

Further practice · Attempt part c only.

10 (a) Using formulae on page 2, show that

(i) $\sin 2A = 2\sin A\cos A$

(ii) $\cos 2A = 2\cos^2 A - 1$

(3)

$$f(\theta) = \frac{2\tan\theta}{1+\tan^2\theta}$$

(b) Show that $f(\theta) = \sin 2\theta$

(4)

(c) Solve, in radians to 3 significant figures, for $-\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$, the equation

$$5\tan\left(x+\frac{\pi}{6}\right) = \left[1+\tan^2\left(x+\frac{\pi}{6}\right)\right]\left[1-2\cos^2\left(x+\frac{\pi}{6}\right)\right]$$

(6)

(d) Using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \left(\frac{4\tan\theta}{1+\tan^2\theta} - \cos 5\theta + 2\right) d\theta$$

(4)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

12. Question 10 · part c · 6 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 7 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

7 (i) (a) Using a formula from page 2, show that

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

(2)

Given that $\tan 2\alpha = 1$

(b) show that $\tan\alpha = a \pm \sqrt{b}$ where a and b are integers whose values need to be found.

(3)

(ii) (a) Using formulae from page 2, show that $\cos(x-30)^\circ = \sin(x+30)^\circ$ can be written as $\tan x^\circ = 1$

(4)

(b) Hence, or otherwise, solve

$$\cos(2y-30)^\circ = \sin(2y+30)^\circ \qquad \text{for } -90 < y \leqslant 90$$

(2)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

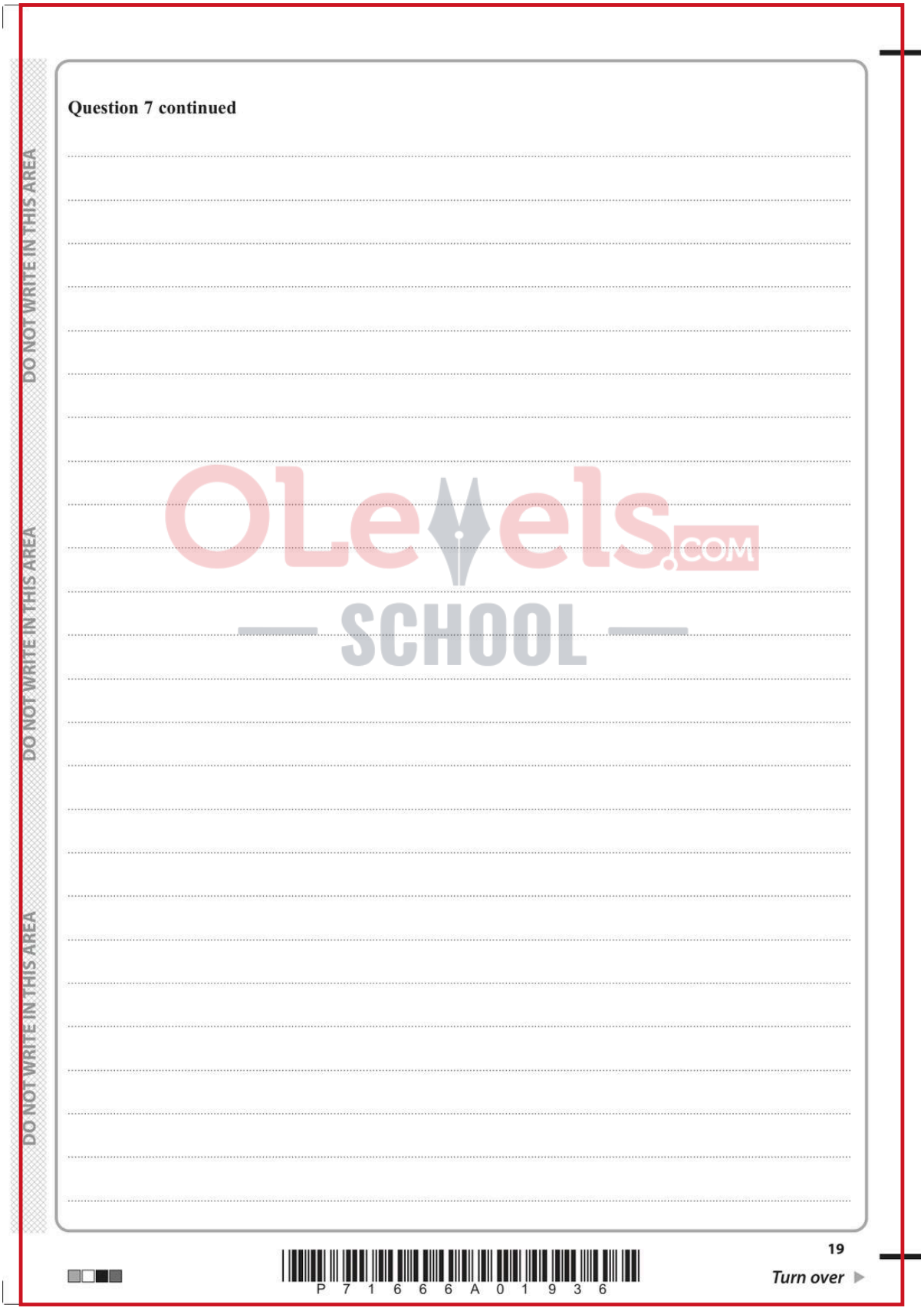
DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

13. Question 7 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.



13. Question 7 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

13. Question 7 · part c · 2 marks

2022 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 7 is 11 marks)

13. Question 7 · part c · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

Practice workbook · Proving identities with double-angle and addition formulae

Rewriting a longer trig expression into a given form by substituting double-angle and addition results, using sin^2 + cos^2 = 1 and tan = sin/cos.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

OLevels.com SCHOOL

Type ID: FPM-TRI-23

1. Question 9 · part b · 5 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part b only.

9 (a) Using the formulae on page 2, show that

(i) $\cos^2 A = \dfrac{\cos 2A + 1}{2}$

(ii) $\sin^2 A = \dfrac{1 - \cos 2A}{2}$

(4)

(b) Show that

$$(2\sin x - \cos x)(\sin x - 3\cos x) = \frac{1}{2}(\cos 2x - 7\sin 2x + 5)$$

(5)

$$y = (2\sin x - \cos x)(\sin x - 3\cos x)$$

(c) Solve, for $0° \leqslant x \leqslant 180°$ the equation, $\dfrac{dy}{dx} = 0$

Give your answers to the nearest whole number.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

1. Question 9 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 10 · part b · 4 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

10 (a) Use the formula for $\cos(A + B)$ to show that $\cos 2A = 2\cos^2 A - 1$

(2)

(b) Show that $\cos 4A = 8\cos^4 A - 8\cos^2 A + 1$

(4)

(c) Solve the equation $\cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right)\left[\cos^2\left(\frac{\theta}{4} + \frac{\pi}{24}\right) - 1\right] = -\frac{1}{16} \qquad 0 \leqslant \theta < 2\pi$

Give your answers in terms of π.

(5)

$$f(A) = 4\cos^4 A - 4\cos^2 A + 1$$

(d) Using calculus, find the exact value of $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} f(A)\,dA$

Give your answer in the form $a\pi - b\sqrt{c}$ where a and b are fractions in their lowest terms and c is a prime number.

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com — SCHOOL —

2. Question 10 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 7 · part a · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

7 Given that

$$\frac{3+\sin^2\theta}{\cos\theta-2}=3\cos\theta$$

(a) show that $\cos\theta=-\frac{1}{2}$

(4)

(b) Hence solve the equation

$$\frac{3+\sin^2 3x}{\cos 3x-2}=3\cos 3x \quad \text{for } 0^\circ \leqslant x < 180^\circ$$

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

P 6 6 3 0 7 A 0 1 8 3 2

3. Question 7 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 11 · part a · 3 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

11 $f(\theta) = (2\cos\theta - \sin\theta)(2\sin\theta + \cos\theta)$

(a) Show that $f(\theta) = \frac{3}{2}\sin 2\theta + 2\cos 2\theta$

(3)

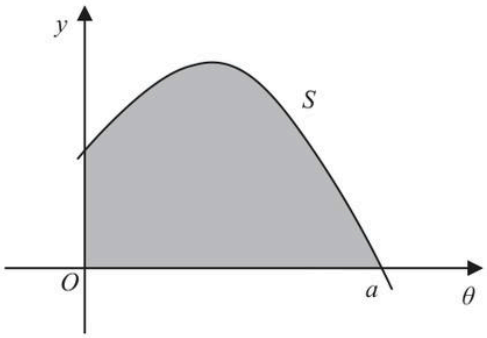


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows part of the curve S with equation $y = f(\theta) + 2$

Given that S intersects with the θ-axis at the point with coordinates $(a, 0)$

(b) using $\sin^2\theta + \cos^2\theta = 1$, or otherwise, show that $a = \frac{\pi}{2}$

(5)

(c) Using algebraic integration, find the exact area bounded by S, the positive θ-axis and the positive y-axis shown shaded in Figure 4

(3)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

4. Question 11 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 10 · part b · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part b only.

10 Using formulae given on page 2

(a) show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

Given that $\theta \neq (90° + 180°n)$ where $n \in \mathbb{Z}$

(b) use the results from part (a) to show that $\sin 2\theta - \tan\theta$ can be written as $\tan\theta\cos 2\theta$

(4)

(c) Solve for $0 < x < 360$

$$\sin 2x° - \tan x° = 0$$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

5. Question 10 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

6. Question 11 · part b · 4 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

11 Using formulae from page 2, show that

(a) (i) $\cos 2A = 2\cos^2 A - 1$ (3)

(ii) $\sin 2A = 2\sin A \cos A$ (1)

(b) Show that $\cos^3 A = \dfrac{\cos 3A + 3\cos A}{4}$ (4)

Hence, or otherwise,

(c) solve, giving exact values in terms of π

$$8\cos^3\left(\frac{\theta}{2}\right) - 6\cos\left(\frac{\theta}{2}\right) - 1 = 0 \qquad \text{for } 0 \leqslant \theta \leqslant 2\pi$$ (4)

(d) use algebraic integration to find the exact value of

$$\int_0^{\frac{\pi}{6}} \left(4\cos^3\theta - \sin 2\theta\right) \mathrm{d}\theta$$ (4)

OLevels.com SCHOOL

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

6. Question 11 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 8 · part b · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

8 (a) Write down the value of k such that $\sin 2A = k \sin A \cos A$

(1)

$$g(A) = 2 + 3\cos A - \sin A - 3\sin 2A - 2\cos^2 A$$

Given that $g(A)$ can be written in the form $(p\cos A - \sin A)(q - r\sin A)$ where p, q and r are integers,

(b) find the value of p, the value of q and the value of r.

(3)

(c) Hence solve, in radians to 3 significant figures where appropriate, the equation

$$g(2\theta) = 0 \quad \text{for} \quad 0 \leqslant \theta < \pi$$

(6)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

7. Question 8 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 1 · part b · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

1 (a) Use the formula for $\cos(A + B)$ to show that $\cos 2A = 1 - 2\sin^2 A$

(2)

Given that $\sin A = \frac{x+1}{2}$ and $\cos 2A = \frac{4-y}{3}$

(b) show that $y = \frac{1}{2}(3x^2 + 6x + 5)$

(3)

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

8. Question 1 · part b · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

Question 1 continued

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

(Total for Question 1 is 5 marks)

8. Question 1 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 11 · part b · 5 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

11 (a) Use a formula on page 2 to show that $\sin^2 A = \frac{1}{2}(1 - \cos 2A)$ (3)

(b) Show that $\sin^4 x + \cos^4 x = \frac{3 + \cos 4x}{4}$ (5)

(c) Hence solve, in degrees to one decimal place, the equation

$$8\sin^4\left(\frac{\theta}{2}\right) + 8\cos^4\left(\frac{\theta}{2}\right) = 5\sin(2\theta) + 6 \quad \text{for} \quad 0^\circ \leqslant \theta < 180^\circ$$

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

9. Question 11 · part b · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 10 · part c · 4 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part c only.

10 (a) Use the factor theorem to show that $(4x - 3)$ is a factor of

$$16x^3 + 11x - 15$$

(2)

(b) Using formulae given on page 2, show that

(i) $\sin 2\theta = 2\sin\theta\cos\theta$

(ii) $\cos 2\theta = 2\cos^2\theta - 1$

(5)

(c) Show that the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

becomes the equation

$$16x^3 + 11x - 15 = 0$$

by using the substitution $x = \cos\theta$

(4)

(d) Hence show that any solution of the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

is given by $\cos\theta = \frac{3}{4}$

(4)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

10. Question 10 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 10 · part b · 4 marks

2023 NOV · Paper 01

Further practice · Attempt part b only.

10 (a) Using formulae on page 2, show that

(i) $\sin 2A = 2\sin A\cos A$

(ii) $\cos 2A = 2\cos^2 A - 1$

(3)

$$f(\theta) = \frac{2\tan\theta}{1+\tan^2\theta}$$

(b) Show that $f(\theta) = \sin 2\theta$

(4)

(c) Solve, in radians to 3 significant figures, for $-\frac{\pi}{2} \leqslant x \leqslant \frac{\pi}{2}$, the equation

$$5\tan\left(x+\frac{\pi}{6}\right) = \left[1+\tan^2\left(x+\frac{\pi}{6}\right)\right]\left[1-2\cos^2\left(x+\frac{\pi}{6}\right)\right]$$

(6)

(d) Using calculus, find the exact value of

$$\int_0^{\frac{\pi}{2}} \left(\frac{4\tan\theta}{1+\tan^2\theta} - \cos 5\theta + 2\right) d\theta$$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM SCHOOL

11. Question 10 · part b · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 11 · part a · 4 marks

2025 JUN · Paper 02

Further practice · Attempt part a only.

11 (a) Show that $\cos^4\theta - \sin^4\theta = \cos 2\theta$

(4)

(b) Hence, or otherwise, solve the equation

$$8\cos^2\left(2\theta + \frac{\pi}{4}\right) - 3 = 2\cos^4\left(\theta + \frac{\pi}{8}\right) - 2\sin^4\left(\theta + \frac{\pi}{8}\right) \quad \text{for} \quad 0 \leqslant \theta < \pi$$

Give your solutions to 2 decimal places.

(7)

(c) Using calculus, find the exact value of $\int_{\frac{\pi}{16}}^{\frac{\pi}{8}} \left(\cos^4 2x - \sin^4 2x - 8\sin 4x\right) \mathrm{d}x$

Give your answer in the form $a - b\sqrt{2}$ where a and b are rational numbers.

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

12. Question 11 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

13. Question 11 · part b · 3 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

11 (a) Express the equation

$$3\sin(A - B) = \sin(A + B)$$

in the form $\tan A = k \tan B$, giving the value of the integer k.

(4)

(b) Given that $\theta \neq \frac{(2n+1)\pi}{2}$ where $n \in \mathbb{Z}$,

show that $\frac{\cos^4\theta - \sin^4\theta}{\cos^2\theta} = 1 - \tan^2\theta$

(3)

(c) Using the exact values of $\sin x°$, $\cos x°$ and $\tan x°$ for $x = 30, 45, 60$

show that

(i) $\cos 15° = \frac{\sqrt{6} + \sqrt{2}}{4}$

(2)

(ii) $\tan 255° = \frac{3 + \sqrt{3}}{3 - \sqrt{3}}$

(4)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.COM — SCHOOL —

13. Question 11 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

14. Question 11 · part b · 3 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

11 (a) Using a formula on page 2 show that $\cos 2A = 2\cos^2 A - 1$

(2)

(b) Hence show that $\left(2\cos^2 A - 1\right)^2 = \dfrac{\cos 4A + 1}{2}$

(3)

The curve with equation $y = \dfrac{\sin 2x}{2} + \dfrac{\left(2\cos^2 x - 1\right)^2}{2} + \dfrac{1}{8}$ has a stationary point P in the range $0 \leqslant x \leqslant \dfrac{\pi}{6}$

(c) Find the exact coordinates of P

(7)

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

DO NOT WRITE IN THIS AREA

OLevels.com SCHOOL

14. Question 11 · part b · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL